

Designing Guardrails to Align LLM-Generated Social Scenarios with the Needs of Autistic Learners

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Background: Autism Spectrum Disorder

- Autism Spectrum Disorder (ASD) is a pervasive developmental disorder characterised by a set of impairments including social communication problems.¹
- ASD has been considered to affect about 1% of the world's population (US Dep. of Health, 2018).²

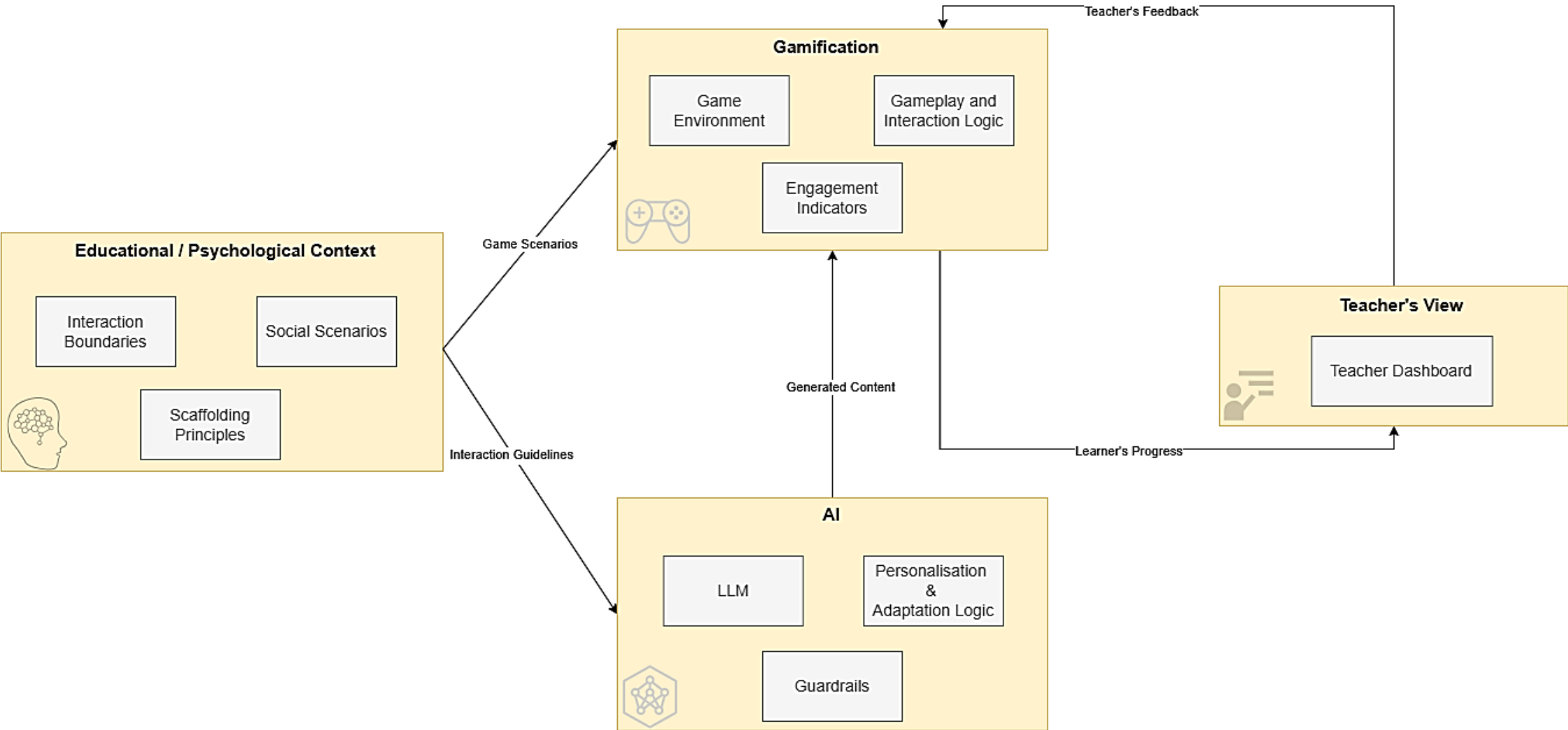
¹ L. Wing, and J. Gould, "Severe Impairments of Social Interaction and Associated Abnormalities in Children: Epidemiology and Classification". Journal of Autism and Developmental Disorders, 9(1), pp.11-29, 1979.

² U.S. Department of Health & Human Services. Data and statistics | autism spectrum disorder (asd) | ncbddd | cdc, 2018. URL: <https://www.cdc.gov/ncbddd/autism/data.html>.

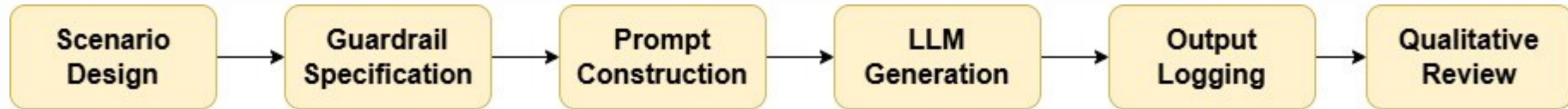
Motivation & Problem

- Social scenarios support autistic learners in practising everyday interactions.
- Scripted scenarios → predictable but limited.
- LLMs → flexible but non-deterministic.
- **Challenge:**
 - How can we keep LLM flexibility while ensuring calm, predictable, and structured interactions?

Project Overview



Methodology Overview



All scenarios were executed using Mistral-7B-Instruct-v0.3.

What “Guardrails” Mean Here

- Explicit constraints embedded in the prompt.
- Define the LLM’s role, objectives, and response limits.
- Enforce:
 - Calm, literal language
 - Short, structured responses
 - Predictable redirection and reassurance
- **Goal:**
Reduce ambiguity and cognitive load while maintaining safe interaction.

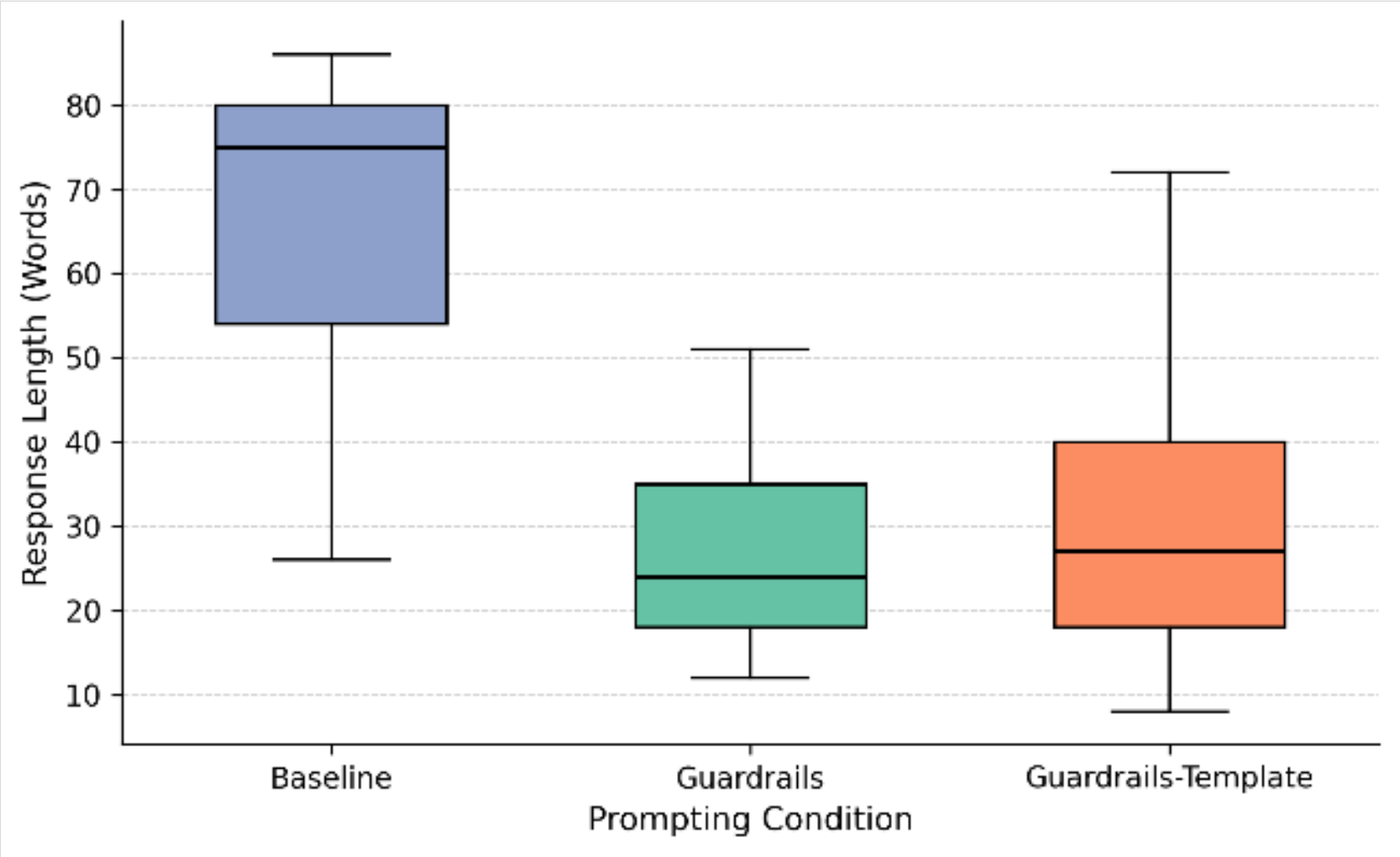
Example: Scenario & Guardrail Template

Scenario Title	Entering the Library and Interacting with the Reception
Context	The player (KS4 or first-year university student) arrives at the campus library to look for documents on green campuses. The librarian is busy helping another student. The player must decide how to behave while waiting.
Assigned Role (LLM)	University Librarian
Prompt Template	You are a university librarian. A student approaches the desk while you are speaking to someone else. Respond kindly, maintaining a professional and calm tone suitable for a 16-year-old. Use short, literal sentences.
Guardrails	• Avoid sarcasm or scolding language. • Keep responses under 3 sentences. • No references to punishment or exclusion. • Always offer a polite redirection ('Please wait a minute, I'll help you next'). • Maintain neutral-positive emotional tone.

Results

- 75 outputs overall.
- 5 scenarios $\times$ 5 user-input variants $\times$ 3 prompting conditions.
- We explored the impact of guardrails on:
 - Tone and Consistency
 - Managing and Resolving Minor Social Misunderstandings
 - Management of Cognitive Load
 - Emotion Recognition as Pedagogical Stimuli

Results (cont'd)



Results: Sample Output

User Input: "Sorry, I'm in a rush. Can you help me quickly?"	
Baseline Output (No Guardrails)	Output with Guardrails
Hi! I understand you're in a hurry. I can help, but I'm currently assisting another student. Could you please wait a moment and let me know what you need once I'm finished?	Please wait a moment. I am helping another student. I will help you next.

Limitations & Conclusions

- The key limitation is using a single model and qualitative analysis only.
- Guardrails may also reduce conversational flexibility.
- However, guardrails increased predictability and tone stability.
- Structured prompts can make LLM dialogue safer for autism-focused social practice.

Thank You!

Please feel free to contact me for any opportunity for collaboration:

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